MATH 1342 – Elementary Statistics – Week 14: Chi-Square Test for Association

Nov 24 – 26 | 7.2 Testing for an Association Between Two Categorical Variables • Thanksgiving Holiday – No Class (Thu Nov 26) | Fall 2026 · Section 036 · TTh 6:00–7:20 PM (DIL)

Module Objectives	Instructional Materials	Learning Activities	Assessments	Tools
1. State the hypotheses for a chi-square test of association from a two-way table. (CLO 8) **2.** Compute expected counts and check that all expected counts are at least 5. (CLO 8) **3.** Compute the chi-square statistic and df = (r − 1)(c − 1), and interpret the right-tail p-value. (CLO 8) **4.** Relate the chi-square test on a 2×2 table to the two-proportion z test, and explain that a significant result shows association but not direction or causation. (CLO 8)	e-Text: 7.2 Testing for an Association Between Two Categorical Variables Interactive Lecture: 7.2 Testing for an Association Between Two Categorical Variables Slides: 7.2 Testing for an Association Between Two Categorical Variables StatKey – Theoretical Distributions (χ^2)	Participation: Chi-Square Test – faculty rank × gender: compute expected counts, cell contributions, χ^2 and df; decide and interpret	Online Homework: 7.2 Testing for an Association Between Two Categorical Variables (due Sun Nov 29, 11:59 PM)	WileyPLUS StatKey Blackboard Zoom (live class)
Course resources and assessments of earlier weeks (not aligned to this week's objectives)			Take-Home Quiz 8 (due Tue Nov 24, 11:59 PM) – covers 6.4, 6.5 Test 3 Retake – Mon Nov 23 – Sun Nov 29 (optional; higher score counts)	Assignment Tool in Blackboard Assessment Tool in Blackboard Respondus LockDown Browser and Monitor

CLO 8 Perform hypothesis testing using statistical methods.